



Paper ID : 260609

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Subject Code: MPH101T

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M PHARM

(SEM. I) THEORY EXAMINATION 2025-26

MODERN PHARMACEUTICAL ANALYTICAL TECHNIQUES

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x 10 = 20

- Define chemical shift with an example.
- Recall the various types of electronic transitions.
- What do you mean by fingerprint region?
- How can you distinguish between a molecular peak and a base peak?
- Define spin-spin coupling.
- Define a fragment ion peak
- Enlist the pharmaceutical applications of HPLC
- Recall the various carrier gases used in gas chromatography.
- What do you glass transition temperature
- Recall the advantages of TGA.

SECTION B

2. Attempt any two of the following:

2 x 10 = 20

- Discuss the principles and instrumentation of flame emission spectroscopy.
- Illustrate the principle and instrumentation of mass spectroscopy.
- Explain the principle and methodology of Affinity chromatography

SECTION C

3. Attempt any five part of the following:

7 x 5 = 35

- Discuss the theory of UV spectroscopy.
- Explain the factors affecting vibrational frequencies of IR spectroscopy.
- Illustrate mass fragmentation and its rules.
- Describe the principle and applications of paper electrophoresis.
- Discuss the principle, instrumentation, and disadvantages of DSC
- Explain the bioluminescence assays.
- Outline the types of crystals and applications of X-ray diffraction.